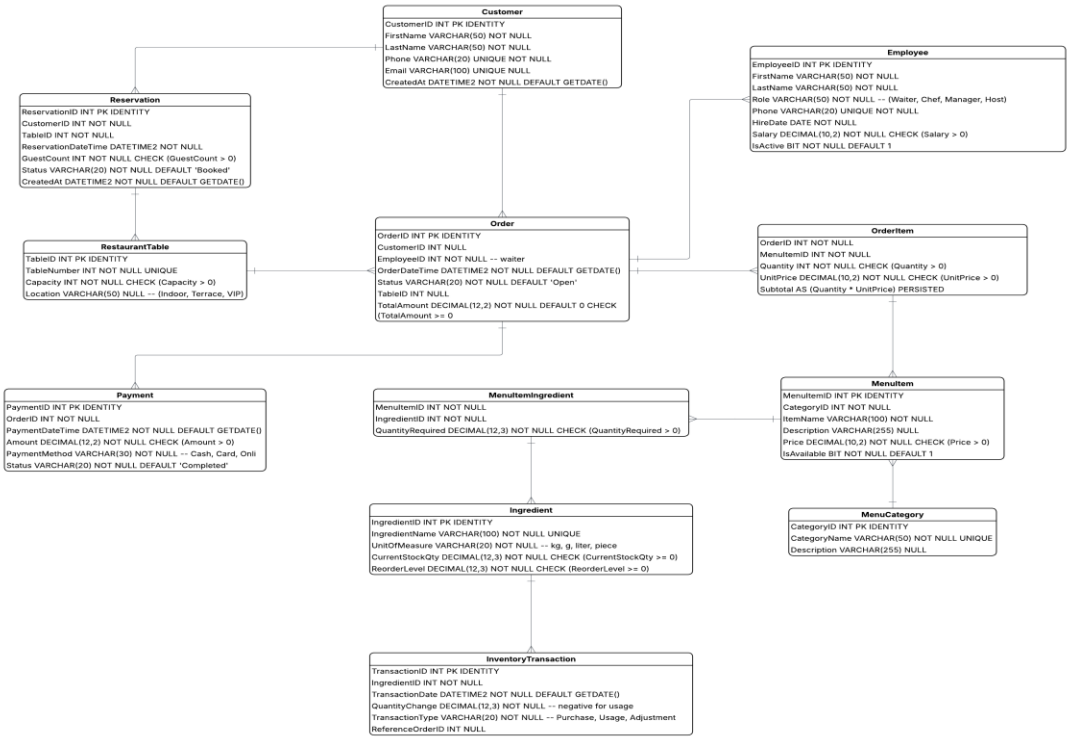
1.1 Logical Scheme



1.2 Objects

1) Customer

Table Description:
Stores information about restaurant customers such as name, phone, and email.

Table Name	Field name	Field Description	Data Type / Constraints
Customer	CustomerID	Unique customer ID	**INT PK**
	FirstName	Customer first name	VARCHAR(50), UNIQUE, NOT NULL

	LastName	Customer last name	VARCHAR(50), NOT NULL
	Phone	Customer phone	VARCHAR(20), UNIQUE
	Email	Customer email	VARCHAR(100)
	CreatedAt	Account creation date	DATETIME2

Comments on table relationships

It stores customer information. One customer can make many reservations.

-Example with data

CustomerID	FirstName	LastName	Phone	Email
1	Aisha	Zhumagali	+77011234567	aisha@mail.com
2	Dias	Ermekov	+77029876543	dias@mail.com

2) **Employee**

Stores information about restaurant employees including their role, salary, and employment status.

Table Name	Field name	Field Description	Data Type / Constraints
Employee	EmployeeID	Unique employee ID	INT, PK, AUTO_INCREMENT, NOT NULL
	first_name	First name	VARCHAR(50), NOT NULL
	last_name	last name	VARCHAR(50), NOT NULL
	Role	Job position	VARCHAR(50) NOT NULL
	HireDate	Hiring date	DATE NOT NULL
	phone	Contact phone number	VARCHAR(20) UNIQUE
	Salary	Employee salary	DECIMAL(10,2) NOT NULL
	IsActive	Active employee	BIT NOT NULL DEFAULT 1

Independent table.
It stores restaurant employee information such as role and salary.

-Example with data

EmployeeID	first_name	last_name	Role	IsActive	Phone	HireDate	Salary
1	Saida	Tauman	Manager	1	+77016667788	250000	1
2	Symbat	Kadyrgali	Waiter	1	+77016667712	120000	2

3) RestaurantTable

Stores information about tables available in the restaurant and their seating capacity.

Table Name	Field name	Field Description	Data Type / Constraints
RestaurantTable	TableID	Table identifier	INT, PK, AUTO_INCREMENT, NOT NULL
	TableNumber	Foreign key refencing the customer	INT NOT NULL UNIQUE
	Capacity	Number of seats	INT NOT NULL CHECK

	Location	Indoor / Terrace / VIP	VARCHAR(50) NOT NULL

Comments on table relationships

Parent of Reservation.

It stores information about restaurant tables. One table can have many reservations at different times.

-Example with data

TableID	TableNumber	Capacity	Location
1	1	4	Indoor
1	2	2	Terrace

4) Reservation

Stores information about table reservations made by customers. It links customers with tables and reservation times.

Table Name	Field name	Field Description	Data Type / Constraints
Reservation	ReservationID	Reservation ID	INT IDENTITY(1,1) PRIMARY KEY
	CustomerID	Unique transaction identifier	INT NOT NULL FOREIGN KEY REFERENCES

	TableID	Reserved table	INT NOT NULL FOREIGN KEY REFERENCES
	ReservationDateTime	Reservation date and time	DATETIME2 NOT NULL
	GuestCount	Number of guests	INT NOT NULL CHECK (GuestCount > 0)
	Status	Reservation status	VARCHAR(20) NOT NULL DEFAULT
	CreatedAt	Created date	DATETIME2 NOT NULL DEFAULT GETDATE()

Comments on table relationships

Child of Customer and RestaurantTable.

It links customers with tables and stores reservation date, number of guests, and status..

-Example with data

ReservationID	CustomerID	TableID	ReservationDateTime	GuestCount	Status	CreatedAt
1	1	13	2026-03-10 19:00	3	Confirmed	2026-03-08

2	5	24	2026-03-10 21:00	2	Pending	2026-03-08

5) Order

Stores information about restaurant orders such as customer, waiter, table number, order date, status, and total amount.

Table Name	Field name	Field Description	Data Type / Constraints
Order	OrderID	Unique order identifier	INT, PK, IDENTITY
	CustomerID	Customer who placed the order	INT, FK
	EmployeeID	Waiter handling the order	INT, NOT NULL
	OrderDateTime	Date and time when the order was created	DATETIME2, DEFAULT GETDATE()
	Status	Current order status	VARCHAR(20), DEFAULT
	TableID	Restaurant table number	INT
	TotalAmount	Total price of the order	DECIMAL(12,2),

Comments on table relationships

One order can contain multiple order items.

-Example with data

OrderID	CustomerID	EmployeeID	OrderDateTime	Status	TableID
1	2	3	2025-03-01 18:30	Open	5
2	1	4	2025-03-01 19:10	Closed	2

6) OrderItem

Stores individual menu items included in each order.

OrderItem	OrderID	ID of the order	INT, FK
	MenuItemID	ID of the ordered menu item	INT, FK
	Quantity	Number of items ordered	INT,
	UnitPrice	Price of one item	DECIMAL(10,2),
	Subtotal	Total price for the item	Computed (Quantity * UnitPrice)

Comments on table relationships

Each order item belongs to one order and references one menu item.

-Example with data

OrderID	MenuItemID	Quantity	UnitPrice
1	4	2	12.00
2	3	1	8.50

7) MenuItem

Stores dishes and drinks available in the restaurant menu..

Table Name	Field name	Field Description	Data Type / Constraints
MenuItem	MenuItemID	Unique menu item identifier	INT, PK, IDENTITY
	CategoryID	Category identifier	INT, FK
	ItemName	Name of the dish or drink	VARCHAR(100), NOT NULL
	Description	Description of the menu item	VARCHAR(255)
	Price	Price of the item	DECIMAL(10,2), CHECK (Price >0)
	IsAvailable	Indicates if the item is available	BIT, DEFAULT 1

Comments on table relationships

Each menu item belongs to a specific category.

-Example with data

MenuItemID	CategoryID	ItemName	Description	Price	IsAvailable
3	1	Caesar Salad	Salad with chicken and parmesan	8.50	1
4	2	Margherita Pizza	Pizza with tomato and cheese	12.00	1

8) InventoryTransaction

Stores records of ingredient stock changes in inventory.

Table Name	Field name	Field Description	Data Type / Constraints
InventoryTransaction	TransactionID	Unique transaction identifier	INT, PK, IDENTITY
	IngredientID	Ingredient related to the transaction	INT, FK
	TransactionDate	Date and time of the transaction	DATETIME2, DEFAULT GETDATE()
	QuantityChange	Change in ingredient quantity	DECIMAL(12,3)
	TransactionType	Type of transaction	VARCHAR(20)
	ReferenceOrderID	Related order ID	INT

Comments on table relationships

Inventory transactions track ingredient usage and stock updates.

-Example with data

TransactionID	IngredientID	TransactionDate	QuantityChange	TransactionType	ReferenceOrderID
1	5	2025-03-01 17:00	-0.5	Usage	1
2	5	2025-03-01 10:00	5.0	Purchase	NULL

9) 'MenuItemIngredient' Table Description

This table shows which ingredients are used in each menu item. It helps to know what is needed for each dish.

Table Name	Field name	Field Description	Data Type
MenuItemIngredient	MenuItemID	Menu item	INT FK NOT NULL
	IngredientID	Ingredient	INT FK NOT NULL
	QuantityRequired	Required amount	DECIMAL NOT NULL

Comments on table relationships

Junction table between MenuItem and Ingredient. Shows which ingredients are used in each menu item, helping manage recipes and inventory accurately.

-Example with data

MenuItemID	IngredientID	QuantityRequired

1	2	0.200
2	5	0.050

10) ‘MenuCategory’Table Description

A catalog of menu categories that organizes menu items into logical groups (e.g., Pizza, Drinks, Desserts) for easier management and reporting.

Table Name	Field name	Field Description	Data Type / Constraints
MenuCategory	CategoryID	Category ID	INT PK IDENTITY
	CategoryName	Category name	VARCHAR(50) NOT NULL UNIQUE
	Description	Category description	INT PK NULL

Comments on table relationships

Parent of MenuItem. It organizes menu items into logical groups (like Pizza, Drinks, Desserts) and ensures items can be tracked without duplicating category data.

-Example with data

CategoryID	CategoryName	Description
1	Appetizers	Starter dishes
2	Desserts	Sweet food

11) 'Ingredient' Table Description

This table shows all ingredients in the kitchen. It shows how much is in stock and the unit (kg, g, liter, piece).

Table Name	Field name	Field Description	Data Type / Constraints
Ingredient	IngredientID	Ingredient ID	INT PK IDENTITY
	IngredientName	Ingredient name	VARCHAR(100) NOT NULL UNIQUE
	UnitOfMeasure	Measurement unit	VARCHAR(20) NOT NULL
	CurrentStockQty	Current stock	DECIMAL NOT NULL
	ReorderLevel	Minimum stock level	DECIMAL NOT NULL

Comments on table relationships

Child of MenuItem (through MenuItemIngredient). It tracks what ingredients are used in each menu item, allowing inventory to be managed separately from the menu items themselv

-Example with data

IngredientID	IngredientName	UnitOfMeasure	CurrentStockQty	ReorderLevel
1	Cheese	kg	12.500	3.000
2	Tomato	kg	20.000	5.000

12) 'Payment'Table Description

A record of customer payments linked to orders, enabling revenue tracking independently of changes to orders or customer profiles.

Table Name	Field name	Field Description	Data Type / Constraints
Payment	PaymentID	Payment ID	INT PK IDENTITY
	OrderID	Related order	INT FK NOT NULL

	PaymentDateTime	Payment time	DATETIME2 NOT NULL
	Amount	Paid amount	DECIMAL(12,2) NOT NULL
	PaymentMethod	Cash / Card / Online	VARCHAR(30) NOT NULL
	Status	Payment status	VARCHAR(20) NOT NULL DEFAULT

Comments on table relationships

Child of Orders (or Members). It serves as a bridge between the customer/order and their financial transactions. This ensures revenue can be tracked independently of changes to the order or customer profile.

-Example with data

PaymentID	OrderID	PaymentDateTime	Amount	PaymentMe
1	101	2026-03-08 12:15:00	1500.00	Card
2	102	2026-03-08 12:30:00	750.00	Cash